School of Engineering

Diploma in AI & Data Engineering

EG2A12

Data Preparation & Visualization

Project Guide

(2026S1)

1. Project Overview:

This ICA is a data analytics project that requires students to apply the CRISP-DM methodology to a real-world problem using public datasets. You will work in teams to analyse data, develop dashboards, and present actionable insights supported by evidence.

Assessment Components:

S/N	Item	Assessment Weightage	Owner	Due
1	**Project**	40%	Team & Individual	Week 17
2	**Project self-reflection journal & peer evaluation**	Not graded, but it is used to support individual assessment in S/N 1.	Individual	Week 17

2. Project Learning Outcomes:

Upon completing this project, you will be able to:

(a) Develop competencies in carrying out data analytics projects by applying CRISP-DM methodology.
(b) Perform data analytics tasks especially in business & data understanding, data preparation, and data visualization by using appropriate software tools and techniques.
(c) Communicate effectively with stakeholders. Present actionable business insights driven by data pattens/trends discovered.

3. Project Scope:

3.1 Background:

Data analytics enables businesses and government agencies to improve operational efficiency, optimise performance, and make evidence-based decisions. This project simulates a real-world analytics workflow, from problem formulation to insight-driven recommendations.

3.2 Scope and Team Structure:

- Students will work in **teams of 3–5**.
- Each team will:
    - Identify a real-world problem or use case.
    - Identify and evaluate relevant public datasets.
    - Perform data preparation, analysis, and visualization.
    - Propose insights and recommendations based on findings.

3.3 Example Project Topics:

(These are broad initial scopes. You may need to narrow down the actual project scope depending on availability of datasets and your business & data understanding.)

- What can be done to achieve the Singapore Green Plan 2030?
- What can be done for the healthcare system to ensure quality medical service for the aging population?
- How to address aging population challenges?
- How to optimise elderly care services?

- What is the best financial planning strategies for retirement for different demographic groups?
- How is the distribution and accessibility of parks and green spaces across different residential areas?
- How can we help industry or residence better management utility (water & energy) consumptions?
- What can we do to improve in waste management?
- Are there sufficient educational institutes and vacancies among regional countries?
- Which university course should students in your course choose after graduation from polytechnic?

Your team will brainstorm and propose a business use case of your choice (not limited to ideas listed above). Topics related to “HDB prices” are not allowed, as they have been extensively studied. **Important!!! Please ensure that there are sufficient appropriate datasets available for you to perform analysis and derive insights on, before you finalize your topic.**

4. Dataset Requirements and Project Proposal Approval

4.1 Dataset Requirements:

- Use at least 3 different datasets for analysis.
- Datasets must be sufficiently large and complex to demonstrate competencies in:
    - Data selection
    - Cleaning and transformation
    - Integration
    - Formatting
- Datasets must support meaningful analysis and insight generation.

You may use websites or generative AI tools to locate public datasets, but please **do NOT artificially create or use generative AI tools to create synthetic data**, as they may not lead to practically relevant insights.

4.2 Project Proposal:

- Please use the project proposal template provided in Brightspace and follow the instructions given. Please do not exceed 3 pages.
- Proposal approval from your **Unit Tutor (UT)** is required.
- Proposal must:
    - Clearly describe the business problem.
    - Specify the datasets to be used.
    - State business and data mining objectives.
- **Submission deadline:** End of **Week 6** (earlier submission encouraged).
- **Not graded**, but feedback will be provided.
- One submission per team.
- **File naming convention:** *EG2A12_A1/2/3/4_<Team X>_proposal.docx*

5. Project Phases (CRISP-DM Aligned):

a. Business understanding -- **Team**
b. Data understanding -- **Team**
c. Initial Data exploration & preparation – **Team**
d. Further data exploration & preparation, Data visualization, Insights & recommendations – **Each student in the team should create at least 1 dashboard page by himself/herself and present the data-driven insights & recommendations for improving the current situation or solving the problem.** Team can have additional combined dashboards. (All individual work must be related to the same project objectives.)
Note: You are allowed to use common or individual datasets. Consolidate all Python codes into one file per team. The PowerBI dashboards can be common or individual files.
e. Discussion (project challenges & mitigation) -- **Team**
f. Conclusion -- **Team**
g. Reference – **Team**

6. Software Requirement:

- **Python**: Mandatory for data preparation and exploration.
- **Power BI**: Mandatory for data preparation and dashboard visualisation.
- Additional tools may be used if justified.

7. Project Final Presentation Requirement:

- Duration: **20 minutes** (15 minutes presentation + 5 minutes Q&A).
- Presentation week: **Week 17**.
- Content should include all the project phases listed in Section 5.
- All team members must present.
- Each student must explain his/her individual dashboard, insights derived and Recommendations/actions proposed.
- Indicate the student's name on the slide(s) showing his/her dashboard.
- Contributions must be clearly indicated at the end of the slides.
- Dress code: **Smart casual**.

8. Project Final Report Requirement:

- Length: **8–12 pages** (excluding cover page and appendices).
- Format: Font: Arial, Size: 12, Line spacing: 1.5, Alignment: Justified
- Cover Page: Include a project title, LU group, team name, names and admission numbers of all team members, and the date.
- The objective of the report is for the team to further elaborate on project phases **c & d** listed in Section 5, as you are unable to cover them in details during presentation due to time limitations.
- Content of the report should include 2 sections:
    i. Data exploration & preparation
    ii. Data visualization, insights and recommendation

Please describe the data exploration & preparation performed and illustrate them using datasets/subsets as examples. Explain the rationale for performing these data preparation steps. Explain the choice of visualization charts. Highlight the data-driven insights & recommended actions.

- You don't need to include other project phases that have been covered during project presentation.
- Include references / data sources links
- Indicate contributions by each team member
- Submission is due by **Week 17**.

9. Project Individual Reflection & Peer Evaluation

- Please use the project individual reflection & peer evaluation template provided in Brightspace and follow the instructions given.
- Not graded, but used to support individual assessment decisions.
- Maximum: **3 pages**.
- One submission per student. It is mandatory.
- **File naming convention:** *EG2A12_A1/2/3/4_<Team X>_<Your Name>_reflection.docx*
- Submission is due by **Week 17.**

10. Submission Instructions (Brightspace):

Required Submissions

Item	Submission Type	Due Week
Project Proposal	Team	Week 6
Final Presentation Slides, Report, Deliverables (Zip Folder)	Team	Week 17
Individual Reflection & Peer Evaluation	Individual	Week 17

- Ensure sufficient time for upload and Turnitin checks.
- For the final project submission All files must be placed into **one ZIP folder**.

  ZIP folder name: *EG2A12_A1/2/3/4_<Team X>.zip*

  Example File Names in ZIP Folder

  - *EG2A12_A1/2/3/4_<Team X>_presentation.pptx*
  - *EG2A12_A1/2/3/4_<Team X>_report.docx*
  - *EG2A12_A1/2/3/4_<Team X>_raw_data_<description>.csv*
  - *EG2A12_A1/2/3/4_<Team X>_processed_data_<description>.csv*
  - *EG2A12_A1/2/3/4_<Team X>_data_prep.ipynb*
  - *EG2A12_A1/2/3/4_<Team X>_visualization.pbix*

11. Team Contribution and Late Submission

- All team members are expected to contribute actively & meaningfully.
- Cases of non-contribution or free-riding must be reported to the unit tutor early for intervention.
- Late submissions will be penalised according to institutional guidelines

12. Assessment Rubrics:

Please refer to the Appendix.

Appendix – Assessment Rubrics for Project (40% Team + Individual, base mark = 100)

Item (Weight)	Task	Task Weight	Poor	Needs Improvement	Satisfactory	Proficient	Excellent
Presentation [Team & Individual] (15)	**P1: Content and organization; Quality of slides [Team]**	**5**	• Presentation content shows no research effort. • Key ideas of the pitch are not clear e.g. how the project is executed, how business insights are derived from visualization and how business objectives and data mining goals are achieved. • The presentation is organised in a haphazard manner. • Main points are absent. • No or irrelevant use of visuals in presenting information.	• Presentation content shows minimum research effort. • Key ideas of the pitch are vague. • The presentation organisation is difficult to follow. • Main points lack elaboration. • Poor use of visuals in presenting information.	• Presentation content shows some research effort. • Key ideas of the pitch lack clarity in some parts. • The presentation is organised in a sequence that can be followed some of the time. • Some of the main points lack elaboration. • Acceptable use of visuals in presenting information.	• Presentation content demonstrates thorough research. • Key ideas are clearly presented. • The presentation is well-organized with a logical flow. • Main points are adequately elaborated. • Effective use of visuals in presenting information.	• Presentation content demonstrates substantial and insightful research. • Key ideas of the pitch are presented very clearly and convincingly. • The presentation is organised in a logical sequence that flows naturally. • Main points are well elaborated. • Slides are interesting & informative and attract audience's attention throughout.
			0 - 1	**2**	**3**	**4**	**5**
	P2: Delivery & command of language [Individual]	**5**	• Speaker appears very uncomfortable and lacks confidence. • Displays poor eye contact, posture, gestures, and vocal expressiveness. • No effort to engage the audience. • Not fluent or poor pronunciation, thus affecting the audience's understanding.	• Speaker appears somewhat uncomfortable and lacks consistent confidence. • Eye contact, posture, gestures, and vocal expressiveness are weak or inconsistent. • Minimal effort is made to engage the audience. • Delivery is hesitant with noticeable errors in pronunciation or fluency, occasionally affecting understanding.	• Speaker appears relatively confident. • Displays reasonably good eye contact, posture, gestures, and vocal expressiveness. • There are some attempts to engage the audience. • Delivery is generally fluent with minor errors in pronunciation or occasional glitches that do not significantly impede understanding.	• Speaker appears confident and comfortable. • Displays effective use of eye contact, posture, gestures, and vocal expressiveness to enhance the message. • Speaker makes a noticeable effort to engage the audience. • Delivery is fluent with clear pronunciation and strong command of language.	• Speaker appears highly confident and engaging. • Displays masterful use of eye contact, posture, gestures, and vocal expressiveness to captivate the audience. • Speaker successfully and actively engages the audience throughout the presentation. • Delivery is polished, articulate, and demonstrates excellent command of language.
			0 - 1	**2**	**3**	**4**	**5**
	P3: Teamwork [Team]	**2**	The presentation is significantly unbalanced, with some members contributing little or none. Teamwork and collaborative spirit are very poor or absent. Transitions between speakers are non-existent or disruptive.	The presentation load is unevenly distributed among team members. Teamwork and collaborative spirit are weak, with limited evidence of cooperation. Transitions between speakers are awkward or poorly executed.	The presentation load is mostly evenly distributed among team members. Teamwork and collaborative spirit are acceptable, with some evidence of cooperation. Transitions between speakers are present but may be somewhat mechanical.	The presentation load is well-distributed, demonstrating good teamwork and collaboration. Transitions between speakers are smooth and logical. Each member contributes meaningfully to the presentation.	The presentation load is seamlessly integrated, showcasing strong teamwork and a highly collaborative spirit. Transitions between speakers are fluid and enhance the overall presentation. Each member's contribution is significant and well-coordinated.
			0	**0.5**	**1**	**1.5**	**2**
	P4: Handling Q&A [Team]	**3**	Unable to understand or address questions with a clear explanation. Demonstrates a very weak thought process and lack of understanding of the project. Responses are irrelevant or absent.	Struggles to understand and address questions adequately. Demonstrates a limited thought process and gaps in understanding the project. Responses are often unclear, incomplete, or hesitant with long pauses.	Able to understand and answer questions to a certain extent, though explanations may lack clarity or depth. Demonstrates a basic thought process with some understanding of the project. Responses may be somewhat hasty or include noticeable pauses.	Responds to questions clearly and thoughtfully, providing adequate explanations. Demonstrates a sound thought process and good understanding of the project, including specific examples where relevant. Responses are prompt and well-articulated.	Provides insightful and comprehensive answers to questions with clear and well-supported explanations. Demonstrates a strong and logical thought process with a deep understanding of the project, using specific and compelling examples. Responses are articulate, concise, and demonstrate excellent command of the subject matter.
			0	**1**	**2**	**2.5**	**3**

Report Writing [Team] (5)	**R1: Content organization and quality of technical writing [Team]**	**5**	• There is no clear chaptering, headings and pagination. • Report is poorly organized with no logical flow. • Significant portion of contents are missing or overlapping. • There are a lot of grammar & spelling errors with no visual aids. • References are irrelevant or missing.	• There are some chaptering, headings and pagination but they are inconsistent. • Content flow is illogical in several sections. • Some of the contents are missing or overlapping. • There are frequent grammar and spelling errors and visual aids of poor quality. • Some of the references are irrelevant or missing.	• Report is organized with chaptering, headings and pagination. • Content flow is not logical for 1-2 chapters. • There are slight missing or overlapping among the chapters. • There are a few grammar & spelling errors. Visual aids are used to aid in the explanation & illustration. • References are relevant and cited partially.	• Report is organized with chaptering, headings and pagination with logical hierarchy. • Content flow is smooth & logical. • There are little missing or overlapping among the chapters. • There are very few grammar & spelling errors. Visual aids are used effectively to aid in the explanation & illustration. • References are relevant and cited.	• Excellent report organization with chaptering, headings and pagination. • Content flow is logical and easy to read and understand. • There is no missing or overlapping among the chapters. • No grammar & spelling errors. Visual aids are use skillfully & strategically. • Reference lists are comprehensive, relevant and cited.
			0 - 1	**2**	**3**	**4**	**5**
Technical Deliverables [Team + Individual] (80)	**T1: Business understanding [Team]**	**5**	• Background is not described. Business objectives are unclear. Current constraints and assumptions made are not assessed. • Data mining goals are not determined. • Project plan is missing or incomplete with many tasks missing. Estimation of efforts is not within reasonable proportion.	• Background vaguely described. Business objectives somewhat unclear. Limited assessment of constraints/assumptions. • Data mining goals are vaguely mentioned. • Project plan is shown with some missing tasks. Effort estimation is significantly disproportionate or very basic.	• Background is described. Business objectives are stated. Current constraints and assumptions made are assessed. • Data mining goals are determined. • Project plan is shown logically with some estimation of efforts, but may lack detail or accuracy.	• Background is clearly described. Business objectives clearly stated. Relevant constraints and assumptions are assessed. • Data mining goals are well-determined. • Project plan is shown logically with reasonable estimation of efforts required for most tasks.	• Background is thoroughly described. Business objectives are well-scoped and practically achievable. Constraints and assumptions made are carefully and comprehensively evaluated. • Data mining goals are well determined and specific enough. • Project plan is shown logically with good estimation of efforts required, demonstrating a strong understanding of task complexity.
			0 - 1	**2**	**3**	**4**	**5**
	T2: Data understanding, initial data preparation & exploration [Team]	**10**	• Considerations in selecting the appropriate data source are not explained. • Data dictionary is missing. • Critical data quality issues are not identified. • Steps in data selection, cleaning, transformation, integration, and formatting, are not performed.	• Considerations in selecting the appropriate data source are vaguely explained. • Data dictionary very brief or contains many errors. • Insignificant number of data quality issues are identified. • Key steps in data selection, cleaning, transformation, integration, and formatting, are not performed or incorrectly performed for the project. All steps are done in Excel, instead of Python. • Justifications of performing these steps are not explained. There is no specific objective observed in performing data exploration.	• Considerations in selecting the appropriate data source are explained at a basic level. • Data dictionary is presented with essential information, although there are some errors. • Some data quality issues are identified through simple data exploration. • Some steps in data selection, cleaning, transformation, integration, and formatting are attempted using Python but may be incomplete, incorrect, or inefficient. • Justifications of performing these steps are provided but lack in depth or clarity.	• Considerations in selecting the appropriate data source are explained. • Data dictionary is presented with essential information. • Key data quality issues are identified and demonstrated. • There is evidence of the key steps in data selection, cleaning, transformation, integration, and formatting performed using Python. • Justifications of performing these steps are explained with reasonable clarity and connection to project goals.	• Considerations in selecting the appropriate data source are well explained with clear rationale. • Data dictionary is correctly presented with all essential information. • All data quality issues are identified through thorough data exploration with evidence shown. • Data selection, cleaning, transformation, integration, and formatting are performed effectively and efficiently using Python demonstrating strong technical proficiency. • Justifications of performing these steps are well explained and thoroughly thought through, demonstrating a strong understanding of the rationale behind each action.
			0 - 2	**3-4**	**5-6**	**7-8**	**9-10**

	T3: Further data preparation & exploration; Data visualizations [Individual]	**15**	• For the data features required for the individual dashboard, no data preparation effort is demonstrated. • Visuals are absent, or entirely inappropriate for the data, or severely misleading. Axes are unlabelled or incorrectly labelled. • The dashboard layout is disorganised and extremely difficult to navigate. Key information is obscured. • No user interactions features (filtering, transition, drilling etc.) implemented. User is unable to retrieve any meaningful visual information.	• For the data features required for the individual dashboard, limited data preparation effort is demonstrated. • Visuals are poorly chosen or implemented, hindering understanding. Labelling is inconsistent or incomplete. • Dashboard layout is confusing and requires significant effort to navigate. Information is not presented logically. • Very limited user interaction features are present and are poorly implemented or ineffective. User struggles to retrieve desired visual information.	• For the data features required for the individual dashboard, some data preparation effort is demonstrated. • Visuals are used effectively to represent the data. Axes are labelled correctly. • The dashboard layout is well-designed and intuitive/easy to navigate. Information is logically organized. • Some user clicking interactions (filtering, transition, drilling, etc.) are built-in. User can retrieve the desired visual information with some effort.	• For the data features required for the individual dashboard, relevant data preparation is performed and several new features are transformed for insights. • Visuals are used effectively following best practices for clarity and impact. Chart types are well-suited to the data and the intended message. • The dashboard layout is professionally designed and aesthetically pleasing. Navigation is easy and requires minimal effort. • Many relevant user clicking interactions are implemented effectively, enhancing data exploration. User can easily retrieve desired visual information.	• For the data features required for the individual dashboard, relevant data preparation is performed effectively and many new features are transformed, demonstrating thorough thought process for data-driven insights. • Visuals are used with exceptional effectiveness and creativity, enhancing understanding and providing insightful perspectives. Demonstrates a strong command of visualization best practices. • The dashboard layout is exceptionally well-designed, impressive, and highly user-friendly. Navigation is seamless and intuitive, demonstrating a strong focus on user experience. • A comprehensive and well-integrated set of user interaction features is implemented, enabling seamless and insightful data exploration. Visual information can be retrieved effortlessly and efficiently.
			0-3	**4-6**	**7-9**	**10-12**	**13-15**
	T4: Business insights & recommendations [Individual]	**15**	• Insights are superficial, generic, and lack connection to the data analysis performed. • Recommendations are irrelevant, or not convincing, or not validated by data.	• Insights are basic and somewhat connected to the data, but lack significant depth or novelty. • Recommendations have limited relevance or a weak connection to the data insights.	• Business insights are meaningful and clearly derived from the data analysis. • Recommendations are relevant and relatively convincing, but some may not be validated by data.	• The insights are in-depth, go beyond the obvious, and demonstrate a strong understanding of the business context. • Recommendations are relevant to the identified insights & the business context, and provided a general direction for action.	• The insights are profound, reveal key strategic opportunities or challenges, and demonstrate exceptional understanding of the business and its data. • Recommendations are relevant and convincing, supported by data evidence and clearly articulated with compelling reasoning. They are highly actionable.
			0-3	**4-6**	**7-9**	**10-12**	**13-15**
	T5: Overall comprehension & application of CRISP-DM Methodology; Quality of technical discussions & sense making [Team]	**5**	• Lack of understanding of the CRISP-DM is observed. • Discussions are minimal or non-existent, indicating a lack of engagement and understanding. Learning points are poorly articulated or absent.	• Shows a basic awareness of the CRISP-DM methodology but demonstrates inconsistent application across different project phases. Understanding of certain phases is unclear or incomplete. • Discussions are superficial and lack depth, indicating limited understanding and engagement with the topics. Learning points are basic and not well-developed.	• Demonstrates understanding of the CRISP-DM methodology and applies it appropriately in most steps of the project. • Discussion demonstrates some degree of understanding of the topics. Learning points are moderately articulated.	• Exhibits a strong understanding of the CRISP-DM methodology and applies it effectively and logically throughout all relevant phases of the project. • Discussion demonstrates good understanding of the topics and critical thinking skills. Learning points are well-articulated and reflect thoughtful reflection.	• Demonstrates a deep and nuanced understanding of the CRISP-DM methodology, applying it strategically and adaptively to the specific project context. • Discussions are in-depth, insightful, and demonstrate strong critical thinking and sense-making. Learning points are thoroughly articulated and demonstrate significant reflection.
			0 - 1	**2**	**3**	**4**	**5**
	T6: Contribution to project ideas and solution development [Individual]	**30**	• Rarely participates in team discussions or remains silent. Offers little to no input in the project ideas and solution development, or actively hinders the process. • Minimal contribution to the project presentation slides & report.	• Participates in team discussions sporadically or inconsistently. Offers limited assistance or input in the project ideas and solution development. • Insignificant contribution to the project presentation slides & report.	• Participates in team discussions most of the time. Contributes to the project ideas and solution development, to some extent, offering suggestions and feedback. • Contribution to the project presentation slides & report are acceptable.	• Consistently and actively participates in team discussions, offering relevant and thoughtful contributions in the project ideas and solution development. • Significant contribution to the project presentation slides & report.	• Actively and consistently leads or significantly shapes team discussions through insightful questions, and the ability to synthesize different viewpoints. Plays a key role in shaping and refining the project ideas and solution development. • Key contributor to the project presentation slides & report.
			0-6	**7-12**	**13-18**	**19-24**	**25-30**
	Total	**100**					